Experiment ‘p_1_8_painting’ Results

June 3, 2026

Experiment outcome: SUCCESS
Bad responses: 0
Responses containing `assume` : 0
Responses containing `expect` : 0
Resolution attempts: 2
Hard fails (resolution): 0
Soft fails (resolution): 0
Verification attempts: 2

Problem Specification

Problem name: p_1_8_painting
Natural language statement: Write a method that returns an imitation of a Piet Mondrian painting. (Search the Internet if you are not familiar with his paintings.) Use character sequences such as @@@ or ::: to indicate different colors, and use - and | to form lines.
Method signature: p_1_8_painting() returns (s: string)

Ensures

- `s == mystring(" !!!@@@@@@@@@@@@@@@@@@@@\n", 5)`

Functional Code Given

```
function mystring(s: string, n: nat): string
{
  if n == 0 then "" else s + mystring(s, n - 1)
}
```

GenAI interactions

Below you will find all interactions between the 'user' (program) and the 'assistant' (OpenAI).

Program → GenAI

```
You are given the following task to perform in Dafny:

Write a method that returns an imitation of a Piet Mondrian painting. (Search the
    ↪ Internet if you are not familiar with his paintings.) Use character sequences such
    ↪  as @@@ or ::: to indicate different colors, and use - and | to form lines.

The signature should be:

p_1_8_painting() returns (s: string)

The method should respect the following contract:

ensures s == mystring(" !!!@@@@@@@@@@@@@@@@@@@@\n", 5)

The contract uses the following dafny code:

function mystring(s: string, n: nat): string
{
  if n == 0 then "" else s + mystring(s, n - 1)
}

These function(s) must not be used in your implementation of the method.

Produce and show only the Dafny body of this method, including the curly braces that
    ↪ surround it. Do not show the signature nor contract. You must not use 'assume' nor
    ↪  'expect' anywhere in your code.
```

GenAI → Program

System fingerprint: null
ID: chatcmpl-DmdM7vPg45Gj0l594erVXTnxziAvK
Model: gpt-5.5-2026-04-23
Created at: 1780483399
Finish reason: stop
Usage: 460 tokens in, and 574 tokens out

```
{
  s := " !!!@@@@@@@@@@@@@@@@@@@@\n !!!@@@@@@@@@@@@@@@@@@@@\n !!!@@@@@@@@@@@@@@@@@@@@\n
      ↪ !!!@@@@@@@@@@@@@@@@@@@@\n !!!@@@@@@@@@@@@@@@@@@@@\n";
}
```

Program → GenAI

```
When using dafny verify, the below error is emitted and verify fails:
```

```
problem_data/p_1_8_painting/program.dfy(8,0): Error: a postcondition could not be proved
    ↪ on this return path
  |
8 | {
  | ^

problem_data/p_1_8_painting/program.dfy(7,11): Related location: this is the
    ↪ postcondition that could not be proved
  |
7 |  ensures s == mystring(" !!!@@@@@@@@@@@@@@@@@@@@\n", 5)
  | ^^


Dafny program verifier finished with 2 verified, 1 error

Correct the error in code by altering only the method body. Produce and show only the
    ↪ Dafny body, including the curly braces that surround it. Do not show the signature
    ↪  nor contract. You must not use 'assume' nor 'expect' anywhere in your code.
```

GenAI → Program

System fingerprint: null
ID: chatcmpl-DmdMPMPmV4zF9UaZxDkRi2GrPrupe
Model: gpt-5.5-2026-04-23
Created at: 1780483417
Finish reason: stop
Usage: 683 tokens in, and 2872 tokens out

```
{
  var line := " !!!@@@@@@@@@@@@@@@@@@@@\n";
  s := line + (line + (line + (line + (line + "")))));

  assert mystring(line, 0) == "";
  assert mystring(line, 1) == line + mystring(line, 0);
  assert mystring(line, 1) == line + "";
  assert mystring(line, 2) == line + mystring(line, 1);
  assert mystring(line, 2) == line + (line + "");
  assert mystring(line, 3) == line + mystring(line, 2);
  assert mystring(line, 3) == line + (line + (line + ""));
  assert mystring(line, 4) == line + mystring(line, 3);
  assert mystring(line, 4) == line + (line + (line + (line + "")));
  assert mystring(line, 5) == line + mystring(line, 4);
  assert mystring(line, 5) == line + (line + (line + (line + (line + ""))));
}
```

Final Program

```
function mystring(s: string, n: nat): string
{
  if n == 0 then "" else s + mystring(s, n - 1)
```

```
}

method p_1_8_painting() returns (s: string)
       ensures s == mystring(" !!!@@@@@@@@@@@@@@@@@@@@\n", 5)
{
  var line := " !!!@@@@@@@@@@@@@@@@@@@@\n";
  s := line + (line + (line + (line + (line + ""))));

  assert mystring(line, 0) == "";
  assert mystring(line, 1) == line + mystring(line, 0);
  assert mystring(line, 1) == line + "";
  assert mystring(line, 2) == line + mystring(line, 1);
  assert mystring(line, 2) == line + (line + "");
  assert mystring(line, 3) == line + mystring(line, 2);
  assert mystring(line, 3) == line + (line + (line + ""));
  assert mystring(line, 4) == line + mystring(line, 3);
  assert mystring(line, 4) == line + (line + (line + (line + "")));
  assert mystring(line, 5) == line + mystring(line, 4);
  assert mystring(line, 5) == line + (line + (line + (line + (line + ""))));
}
```

Total Token Usage

Input tokens: 1143
Output tokens: 3446
Reasoning tokens: 3072
Sum of 'total tokens': 4589

Experiment Timings

Overall Experiment started at 1780483398699, ended at 1780483484370, lasting 85671ms (85.67 seconds)
Iteration #1 started at 1780483398699, ended at 1780483417349, lasting 18650ms (18.65 seconds)
Iteration #2 started at 1780483417349, ended at 1780483484370, lasting 67021ms (67.02 seconds)